Geospatial Analytics Agents using Large Action Models

Abstract

This project aims at disrupting the traditional processes of contract generation by leveraging OpenAI's GPT-3. 5 Turbo(OpenAI, 2023) using data manipulation and visualization tools including PANDAS, GEO-PANDAS, and FLASK to design a system that would allow the user to ask questions in a simple natural language and provide answers in the form of a map with data points of interest indicated. The aim is to leverage the opportunity of the greater natural language understanding of LLMs to accurately analyze, parse and respond to user inputs in context via proper explanations and query simplification as a way of building and enhancing common data interfaces that do not require extensive technical know-how to use.

The methodology begins with the user inputting a query through a web interface. This query is then refined using predefined system prompts that enhance its clarity and relevance based on the dataset being analyzed. The refinement process ensures that the query is contextually accurate and better understood by the LLM (OpenAI, 2023). Next, the refined query, along with the original input, is processed by GPT-3.5 Turbo. The LLM generates a detailed response, guided by system prompts that ensure precision and contextual appropriateness.

Following this, the response from the LLM undergoes further processing using Pandas for data manipulation. Pandas is utilized for its robust capabilities in handling, filtering, and aggregating data, which are crucial for performing the necessary computations (McKinney, 2010). GeoPandas extends these capabilities to geospatial data, allowing for sophisticated geospatial analysis and manipulation (Jordahl et al., 2014).

The results from the data processing steps are then visualized. Flask, a lightweight web framework, serves as the backbone of the web application, providing a user-friendly interface for interacting with the system (Grinberg, 2018). HTML, CSS, and JavaScript are employed to build a responsive and intuitive front-end. For map visualizations, Folium, a powerful library that leverages the mapping capabilities of Leaflet.js, is used to create interactive maps that display geospatial data